

INTERNATIONAL STANDARD



**Liquid crystal display devices –
Part 40-6: Mechanical testing of display cover glass for mobile devices –
Retained biaxial flexural strength (abraded ring-on-ring)**



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

LIQUID CRYSTAL DISPLAY DEVICES –

**Part 40-6: Mechanical testing of display cover glass for mobile devices –
Retained biaxial flexural strength (abraded ring-on-ring)**

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International Standard IEC 61747-40-6 has been prepared by technical committee 110: Electronic display devices.

The text of this International Standard is based on the following documents:

CDV	Report on voting
110/882/CDV	110/929A/RVC

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 61747 series, published under the general title *Liquid crystal display devices*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
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INTRODUCTION

Mobile electronic devices have become increasingly sophisticated and often incorporate displays for the purposes of user interface and viewing. Such displays commonly incorporate a transparent cover glass which aids in protecting the display against the introduction of damage through routine device transport and use, as well as occasional or accidental misuse.

The purpose of this document is to provide mechanical testing procedures for cover glasses utilized in such applications. Such glasses can be strengthened, for example via an ion-exchange process, which acts to increase mechanical strength through the introduction of a surface compressive layer.

LIQUID CRYSTAL DISPLAY DEVICES –

Part 40-6: Mechanical testing of display cover glass for mobile devices – Retained biaxial flexural strength (abraded ring-on-ring)

1 Scope

This part of IEC 61747 is a mechanical performance testing procedure for cover glass used in electronic flat panel displays in mobile devices. This document focuses on the measurement of surface fracture load after flaw introduction via grit particle abrasion. After abrasion, the retained surface fracture load is measured with the method documented in IEC 61747-40-4.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61747-40-1, *Liquid crystal display devices – Part 40-1: Mechanical testing of display cover glass for mobile devices – Guidelines*

IEC 61747-40-4, *Liquid crystal display devices – Part 40-4: Mechanical testing of display cover glass for mobile devices – Biaxial flexural strength (ring-on-ring)*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1

specimen

individual piece of glass to be abraded and then tested for retained fracture load

3.2

sample

group of specimens that share a common pedigree (such as manufacturing process and period of production), for which failure statistics can be generated and reported

3.3

sample size

number of specimens in a sample

3.4

specimen holder

two-part fixture that consists of a recessed cutout for the specimen and a backing plate to hold the specimen in place when testing

3.5

go/no go gauge

gauge used to ensure the specimen holder hole is within specified tolerances

4 General

This test is performed in two steps:

- 1) controlled abrasion of the surface of a number of specimens in the sample;
- 2) retained failure load testing of the abraded specimens using the ring-on-ring test described in IEC 61747-40-4, with the abrasion damage placed in tension during the test.

This document describes the apparatus and procedures for doing the controlled abrasions. It describes the retained failure load testing referenced in IEC 61747-40-4.

The specimens to be tested are typically 50 mm in width and 50 mm in length, with thickness ranging from 0,55 mm to 2,0 mm. The sample size is 30, as an example. The Weibull statistics [1]¹ of fracture loads, after abrasion, are reported.

The abrasion process introduces flaws to the glass surface by fine particle grit abrasion. Before abrasion, a specified amount of abrasion material is prepared. The specimen is then placed in the specimen holder, with the desired abrasion surface oriented towards the particle containment tube of the device, and inserted under the containment tube. The compressed air is turned on for the device, and the abrasion material is quickly loaded into the abrasion device to abrade the specimen. After the specified abrasion time, the specimen holder is removed from the device and the residual abrasion material is removed from the specimen, using pressurized air, before storage for the allotted inspection time to ensure no premature failures occur to the specimens.

Abrasion pressures are generally 34,5 kPa to 310,3 kPa, more typically 34,5 kPa to 103,4 kPa, and rarely exceed 413,7 kPa. However, selected pressure will vary depending on the damage resistance and thickness of the glass specimens under test. An iterative process, in which a small sample set (10 specimens) is tested at an initial abrasion pressure, and check (i.e. flaw) depths are measured via fractography, may need to be employed to determine the appropriate abrasion pressure that generates check depths similar to those experienced in the field for a given glass composition or condition. Information on the measurement of check depths is described in Annex A. Once an appropriate abrasion pressure is determined for a glass, that pressure may then be used to generate a relative comparison with the performance of other glass compositions/candidates at the same pressure. The general requirements of the abrasion apparatus are given in Clause 5. Apparatus dimensions can be affected by specimen dimensions.

Abrasion testing procedures are given in Clause 6, while retained failure load testing procedures are given in Clause 7 via reference to IEC 61747-40-4.

This test measures the retained surface fracture load by forcing a ring through a specimen that is supported by another, larger ring. The two rings shall be parallel and concentric and the applied force shall be perpendicular to the surface formed by the top of the support ring. The loads at fracture are measured. These loads are not normalized to strength using factors such as specimen dimensions and material properties because significant non-linearities can exist that render the classical formulas inaccurate.

¹ Numbers in square brackets refer to the Bibliography.

The test is applied to a sample of several specimens. The sample statistics of the fracture load values are defined in IEC 61747-40-4. The statistical values to be reported or specified are given in IEC 61747-40-4.

The specimen is placed on the support ring so that it is centered on the ring. Before the specimen is placed on the support ring, the surface that will contact the load ring is covered with a layer of polymeric adhesive tape to preserve the fracture surface and reduce the scattering of glass fragments upon breakage. The support ring is also covered with PTFE film to minimize contact damage and friction.

It is assumed that all measurements are performed by personnel skilled in the general art of mechanical property measurements. Furthermore, it shall be assured that all equipment is suitably calibrated as is known to skilled personnel and that records of the calibration data and traceability are kept.

5 Apparatus

5.1 Testing environment and pre-conditioning

The standard testing environment is specified in IEC 61747-40-1. Specimens shall be stored in such an environment for at least 4 h before and 12 h after abrasion, as well as 24 h before failure load testing.

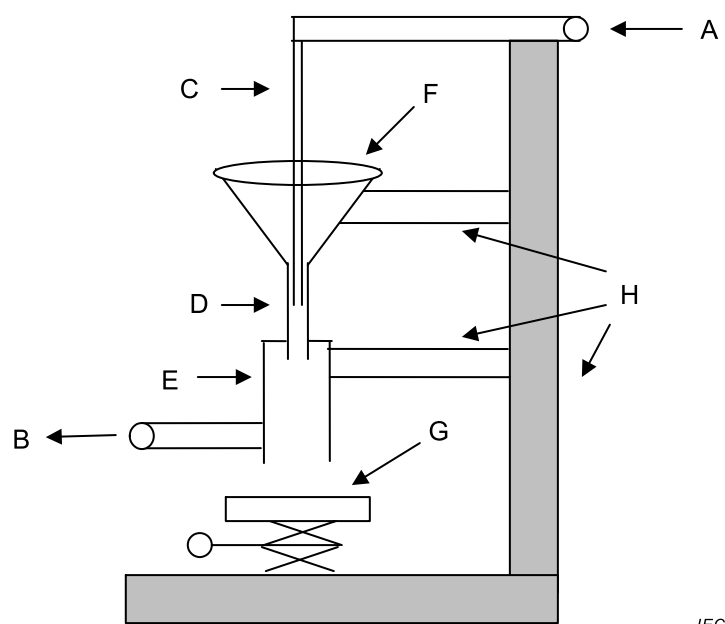
5.2 Abrasion device

The abrasion device is illustrated in Figure 1. The main elements include a funnel for the introduction of particulate into an air stream that impels the particulate onto the surface of the specimens. The air stream is defined in terms of an input air flow at a defined pressure together with three concentric tubes, in order:

- 1) pressurization tube
- 2) particulate acceleration tube
- 3) containment tube

The containment tube has an output exhaust tube to remove most of the particulate after impacting the specimen.

During the abrasion process, the specimen holder (Figure 2) is maintained flush with the bottom of the containment tube such that all air flows are defined within the apparatus. This will minimize the amount of particulate discharged into the laboratory environment. To further prevent air borne particulate, additional housing may optionally (not shown) be provided with a separate exhaust system.

**Key**

A Input pressurized air

B Output exhaust air

C Pressurization tube

D Acceleration tube

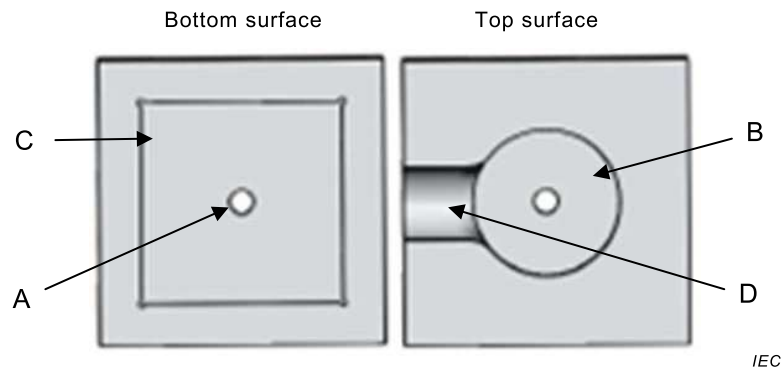
E Containment tube

F Funnel

G Specimen holder support and elevation mechanism (optional)

H Mechanical supports

Dimensions:Pressurization tube inside diameter: 4,8 mm $\pm$ 0,025 mmAcceleration tube inside diameter: 8 mm $\pm$ 0,025 mmContainment tube inside diameter: 32 mm $\pm$ 0,025 mmFunnel mouth diameter and height: 66 mm $\pm$ 0,025 mm and 51 mm $\pm$ 0,025 mmAcceleration tube length: 51 mm $\pm$ 1 mmDistance from bottom of acceleration tube to bottom of containment tube: 28 mm $\pm$ 1 mmDistance from bottom of containment tube to top of sample: 2 mm $\pm$ 0,5 mmDistance from bottom of acceleration tube to bottom of containment tube: 51 mm $\pm$ 1 mmInput pressurized air: $\geq 34,5$ kPa, $\leq 413,7$ kPa $\pm$ 3,4 kPaOutput exhaust vacuum: 24,9 mbar $\pm$ 2,5 mbar**Figure 1 – Abrasion device**



Key

- A Abrasion orifice
- B Containment tube insertion orifice
- C Specimen cutout
- D Output exhaust cavity

Figure 2 – Specimen holder

5.3 Abrasion material

Abrasion material: 90 grit silicon carbide (SiC).

Quantity: 1 cm³ for each specimen.

5.4 Pressurization source

Pressure provided by clean, compressed in-house or portable air source.

6 Procedure

6.1 Safety

6.1.1 Hazard – Broken glass

Wear safety glasses with side shields at all times. Wear gloves when handling broken glass.

6.1.2 Hazard – Airborne particulate

Wear positive pressure air mask or other form of respirator if the unit is not fully enclosed with exhaust.

6.2 Sample

The sample size is 30, excluding any specimens rejected for pre-existing damage.

The specimens of a sample should be provided in a container that supports the specimens on their edges and prevents specimens from contacting one another to prevent transport damage. When the abrasion testing is completed for the specimens of the sample, the same container should be used to send them to the strength testing laboratory.

When the abraded specimens are placed in the container to be sent to the strength testing laboratory, always place them into the container with a fixed orientation relative to the side that is abraded. This orientation should be indicated on the container so that the strength tester can orient the specimen correctly for the strength test.

Verify that the specimen holder is sized correctly for the specimens.

Record any exceptions to the standard procedure. These exceptions can be part of the instructions associated with some experimentation, for example.

6.3 Abrasion testing – Individual specimen

For each specimen:

- a) verify the hole dimensions in the specimen holder by placing the go/no go gauge into the specimen holder. If the go/no go gauge inserts fully with little to no applied pressure, the fixture is worn and shall be replaced;
- b) remove the specimen from the container and place it into the specimen holder recess with the backing plate place on top of the specimen;
- c) prepare the abrasion material by filling a level 1 cm³ scoop;
- d) turn on the input air pump and vacuum exhaust;
- e) move the specimen holder under the containment tube. The specimen holder should be held flush and centered to the bottom of the tube;
- f) dump the scoop into the funnel quickly (within 1 s);
- g) run the input air pump and vacuum exhaust 5 s after the particulate is dumped;
- h) remove the specimen from the specimen holder;
- i) remove the residual abrasion material using pressurized air;
- j) place the specimen into the return container with the abraded side oriented correctly;
- k) allow the specimens to sit in the controlled environment for at least 12 h after abrasion, then re-inspect for failure.

6.4 Complete the report

The report shall include:

- number of sample failures during abrasion and after a 12 h or more hold
- input air pressure used.

Individual customers may require additional information.

7 Retained failure load testing

After the 12 h and more hold and re-inspection, complete the retained failure load testing according to IEC 61747-40-4 with the abraded sides of the specimens placed in tension during the test.

8 Specifications

The following testing aspects require specification to obtain comparable results:

For the abrasion process,

- the aspects listed in IEC 61747-40-4, including sample size and specimen dimensions
- input air pressure
- abrasion material and grit size
- abrasion material dose size
- abrasion time

Annex A (informative)

Check depth measurement

The measurement of check depths created via field failures or from particle abrasion can be conducted using an optical microscope and image analysis software. The check depth is defined as the vertical length of the check/flaw at the failure origin created during damage introduction. Figure A.1 illustrates a typical semicircular glass fracture surface in which the critical flaw is located on the surface. The check depth a is measured from the glass surface to the end of the check and/or median crack. Examples of check depth measurements in glass are shown in Figure A.2 to A.5 and in Lawn [2].

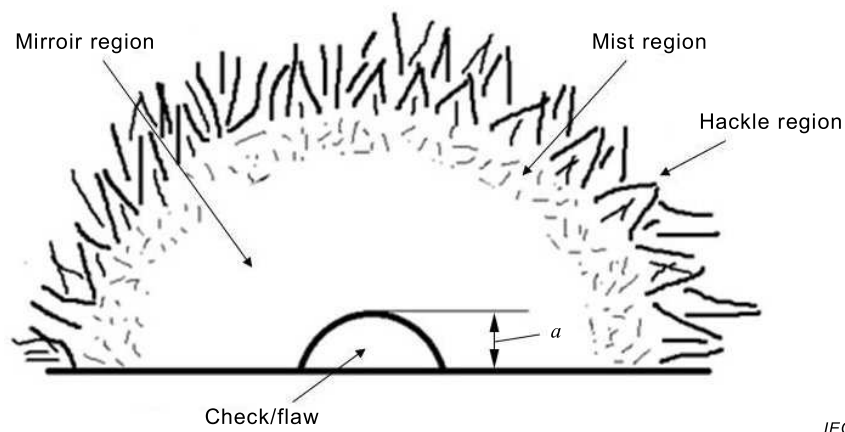


Figure A.1 – Schematic of fracture surface with measurement of check depth

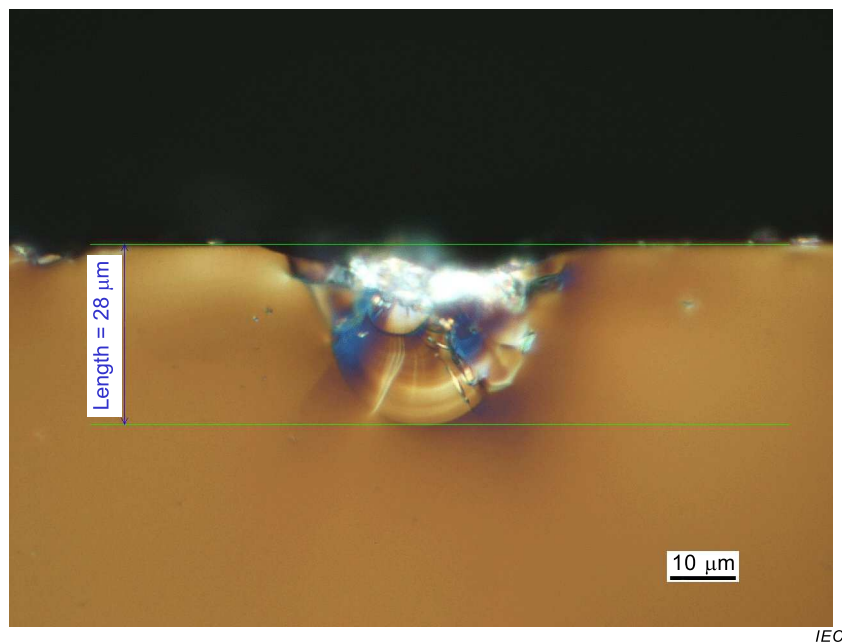


Figure A.2 – Check depth measurement for grit particle abraded glass – Example 1

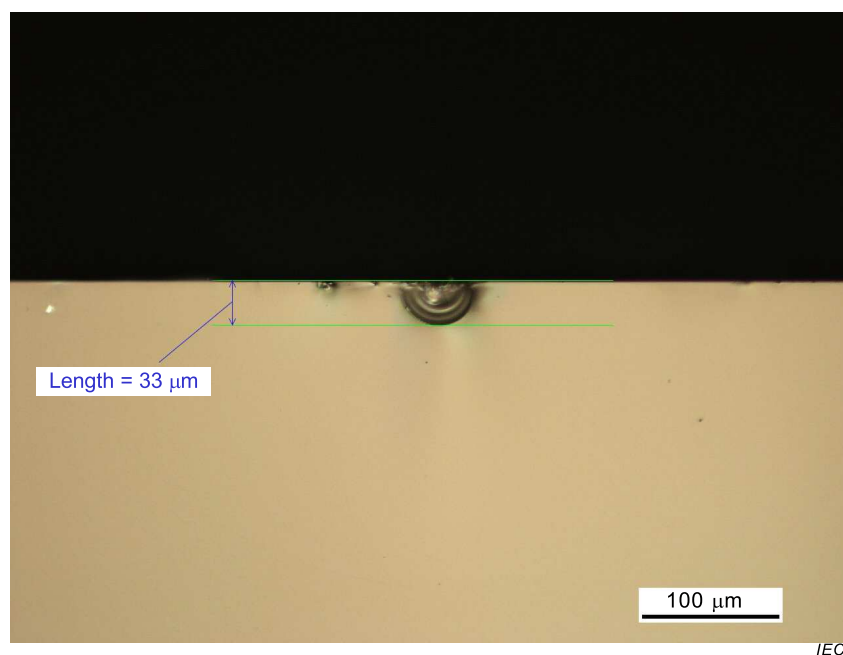


Figure A.3 – Check depth measurement for grit particle abraded glass – Example 2

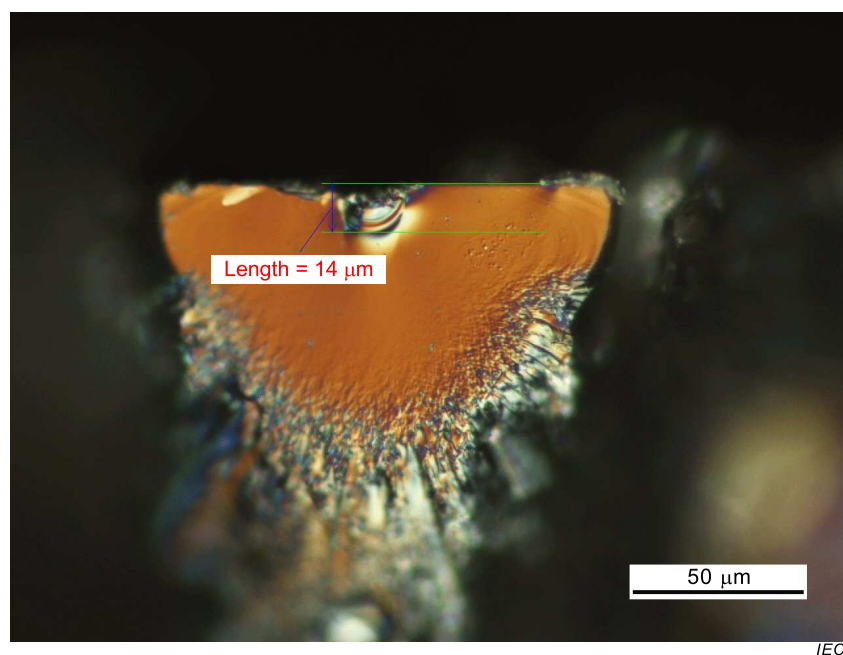


Figure A.4 – Check depth measurement for grit particle abraded glass – Example 3

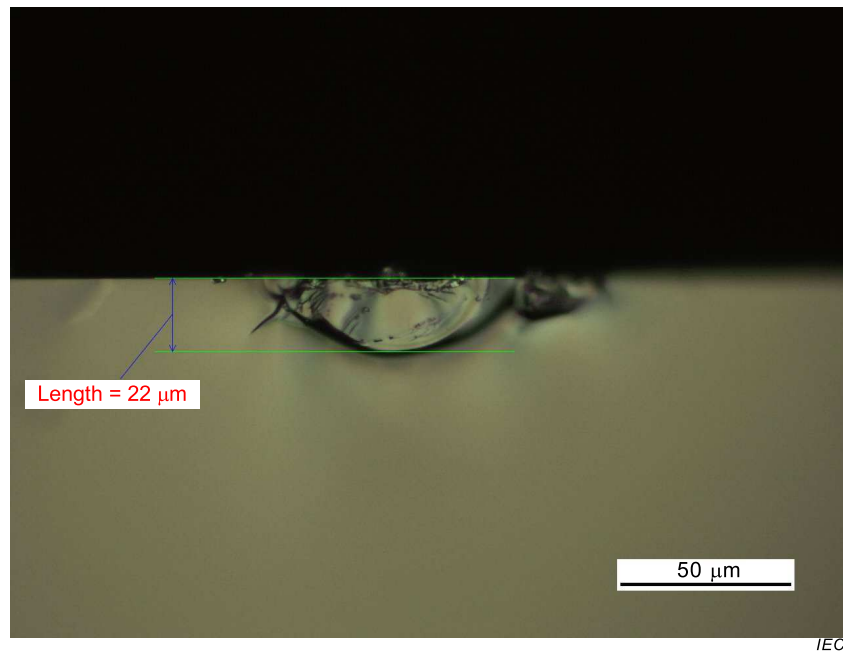


Figure A.5 –Check depth measurement for grit particle abraded glass – Example 4

Bibliography

- [1] IEC 61649, *Weibull analysis*
 - [2] Lawn, Brian, *Fracture of Brittle Solids*, Cambridge: Cambridge University Press, 1993
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